Activity 8 Guide

Course: Semiconductor Process Fundamentals Masterclass

 Course code: C716
 Package: v1.0 | 1 October 2026
 Activity: 8 - Run a Future-Technology Integration Gate
 Alignment: LO2
 Duration: 50 minutes

Goal

Combine the entire course into a structured technology-gate recommendation without treating future devices as isolated marketing concepts.

Scenario

A product team is deciding whether to adopt chiplet packaging, FinFET or GAAFET for its next platform. Integration readiness must be assessed using process, performance, reliability and packaging evidence.

What You Need

- Technology-gate worksheet and evidence template
- Legacy packaging, heterogeneous integration, chiplet, FinFET, GAAFET, vertical transistor and 2D-material slides
- Outputs from Activities 1-7

Detailed Step-by-Step Procedure

1. Choose one candidate: chiplet integration, FinFET scaling or GAAFET scaling.
2. List the required process modules and identify which existing modules need a new material, geometry, tool or control method.
3. Map three critical cross-module interactions and the performance metric each can affect.
4. Define device metrics (for example leakage, drive current, resistance or capacitance) and reliability metrics relevant to the candidate.
5. Add packaging/interconnect implications, including thermal, mechanical and known-good-die considerations where applicable.
6. Create a five-row evidence gate: process capability, metrology readiness, electrical performance, reliability and yield/manufacturability.
7. Set a pass/conditional/fail threshold for every gate.
8. Identify the largest uncertainty and design the smallest engineering experiment that can reduce it.
9. Present a recommendation: proceed, proceed with conditions, or defer.
10. Record the next owner, due date and evidence required to close the decision.

Acceptance Criteria

[ ] The recommendation integrates process, device, reliability, yield and packaging evidence.

[ ] Every gate has a measurable threshold and an accountable follow-up action.

[ ] The conclusion is conditional on evidence rather than technology hype.

Reflection Questions

1. Which legacy process controls remain useful for the chosen technology?
2. What new metrology or reliability evidence is essential before volume manufacturing?

Evidence to Submit

- Completed worksheet.
- Completed evidence template with file names or screenshots.
- The activity output named with the A08- prefix.
- A final release/hold/recommendation statement supported by measurements.

Troubleshooting

- If a workbook opens in protected view, save a local copy before editing blue input cells.
- If formulas do not update, enable automatic calculation and press Ctrl+Shift+F9 (Windows) or Command+= (macOS Excel).
- If a diagram becomes crowded, separate the process flow from the evidence/control loop; cross-reference them with numbered hand-offs.
- If the conclusion is uncertain, state what is known, what is not known, and the next measurement that can resolve it.